

Sleeping Machines

Deep learning that computes with time

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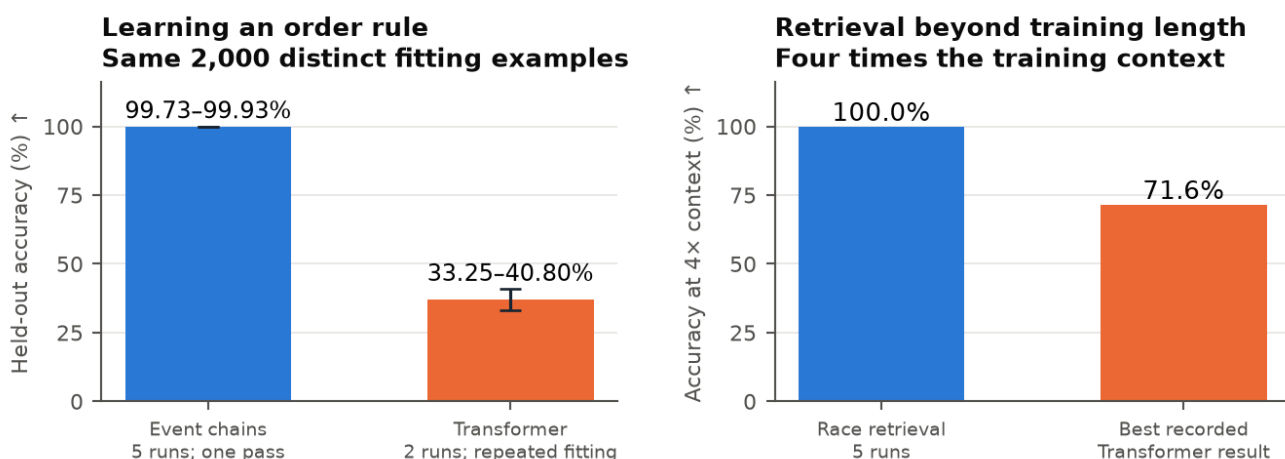
Messages carry a learned vector and an arrival time. Nodes accumulate local memory, transform messages and compete through learned delays. Arrival order and winning races determine the computation. The goal is useful intelligence with much less active work.

The differentiators at a glance

- **Time performs computation.** Delays, races and phase transformations implement useful functions.
- **Hard routes can learn.** Winning messages execute; unrealized alternatives receive counterfactual credit.
- **Deep, persistent event representations.** Vector messages and local memory carry information and credit through layers; retrieval and temporal primitives share the model family.
- **Capacity beyond activity.** The scaling goal is more useful dormant capacity, selectively recruited and judged by prediction quality at a given total work budget.

The strongest demonstrated results

- **Generalization.** Race retrieval reaches **100% at four times the training context** within 4,000 examples in all five runs. A learned phase rule solves **all 3,440 unseen modular triples**, using the supplied period 17.
- **Learning from fewer examples.** Depth-three event chains reach **99.73-99.93%** after 2,000 examples seen once; saved Transformer controls reach **33.25-40.80%** with the same number of distinct examples and repeated fitting. Depth-four chains reach 99.9-100%.
- **Learned representations.** The persistent language model reaches **3.351 development bpc** in a small screen. A learned speech encoder reaches **79.69%** on 512 private development utterances. These models learn source embeddings, temporal state and vector maps.



Left: means and recorded ranges, five event runs and two Transformer runs; 2,000 distinct examples, seen once / presented 400,000 times. Right: all five event runs reach 100% within 4,000 examples; the control is the best saved result across seven Transformer configurations and their learning curves. These synthetic tasks use different architectures and structural priors. Sources: E53/E36 and E61.

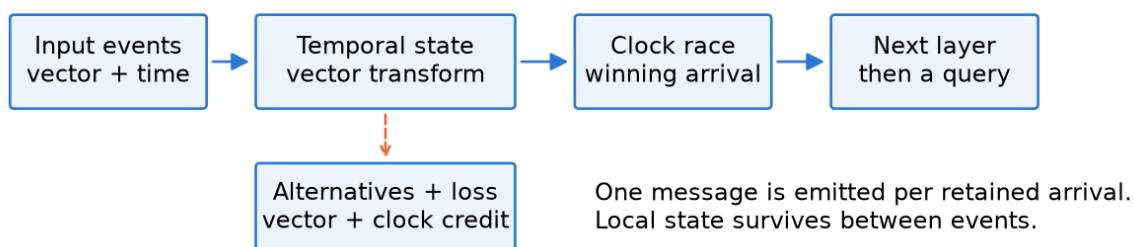
Language scale-up: the 1.21M-parameter learned model is running on 10M text8 characters. Its test result is pending; completed neural controls and cost protocols are in Appendix B.

One architecture, several learned computations

The common idea is local computation triggered by an arrival: retain memory, combine the incoming vector with that memory, and choose an outgoing time. A delay changes which messages meet and which race finishes first. This makes timing part of the learned function. The model family implements this idea at several levels of generality.

A learned event-state block

Source identity, content and elapsed time determine the next vector and arrival.



Model	Mechanism	Evidence	What it establishes
Learned event-state encoders	Source embeddings, temporal modes, nonlinear vector maps and competing clocks	Language E176; speech E165	Learned representations and persistent state
Routed event query encoders	Candidate payloads, receiver memory and hard value/time races	Breadth E120; language E133	Trainable event depth across tasks; text/market breadth variants add statistical evidence
Structured event mechanisms	Temporal chains, relative pointers and phase composition	E34/E53/E54, E61, E124	Sample efficiency and generalization with declared structural priors
Statistical comparison systems	Conditional counts, backoff and copy probabilities	E173; online pilot	Separate baselines; no event backbone

Each task has separately fitted weights. Current learned encoders use fixed depth and locally dense vector maps; the language scheduler retains state and delayed messages across chunks. The routed query encoder instead rebuilds a supplied context per query. Clock learning uses declared surrogate credit through a hard schedule; it is not an exact derivative of every order change.

How the model computes and learns

Time performs computation. An arrival time is a computational value. A delay adds to that value; a first-arrival race computes a minimum and selects a payload; coincidence detects the latest required arrival. Inhibition can veto a path. With a declared periodic reference, phase transformations compose reusable modular relations. Learned timing therefore changes the function and its causal paths, beyond deciding when a fixed dense calculation runs.

- **Local temporal memory** accumulates observed content and elapsed time without evaluating empty time ticks.
- **Computation through delays** uses waiting times, arrival order and clock races to transform information and select outcomes.
- **Hard races and counterfactual credit** choose the emitted vector and delay. Losing alternatives teach better routes while remaining distinct from the winning forward message.
- **Separate keys and values** let content-dependent keys set routes and clocks while value learning preserves the selected schedule.
- **Trainable depth** carries representations and learning credit through a hierarchy. The reversible construction preserves conditional norms under its stated assumptions; readout visibility and routing remain necessary.
- **Structured memories** include relative pointers, learned phase transformations and conditional evidence. Their task gains retain their declared priors; statistical experts are labeled separately.
- **Appropriate supervision** teaches a completed class decision or the next event's type and waiting time, including information carried by silence.

For a temporal pattern such as A followed by B, a learned delay can bring A's trace into coincidence with B. A competing path can veto the match when C intervenes. The timing-pattern and compositional experiments test these mechanisms; the language and speech encoders learn richer vector messages and temporal state.

The primitives and their symmetry limits are developed in temporal computation theory, §56; counterfactual learning, key/value separation and reversible depth give the learning contracts. Clockless delay/race networks compute relative timing relations; phase arithmetic requires its reference. Each implemented model uses a declared subset. Candidate discovery and training alternatives are charged to the work ledger.

A demonstrated advantage: generalization with less work

The retrieval and phase computations preserve useful rules when the evaluation extends beyond the fitting examples. The saved compact Transformer and LSTM controls use the same synthetic evaluation targets. **Higher and further left is better:** more accurate answers from less estimated inference work. All points use one logical operation ledger.



Event computation	Held-out capability	Estimated work per query
Periodic path; 69 learned scalars	3,440/3,440 unseen triples	188 logical operations
Two-layer carrier + hard pointer	100% at four times context	728,602 logical operations

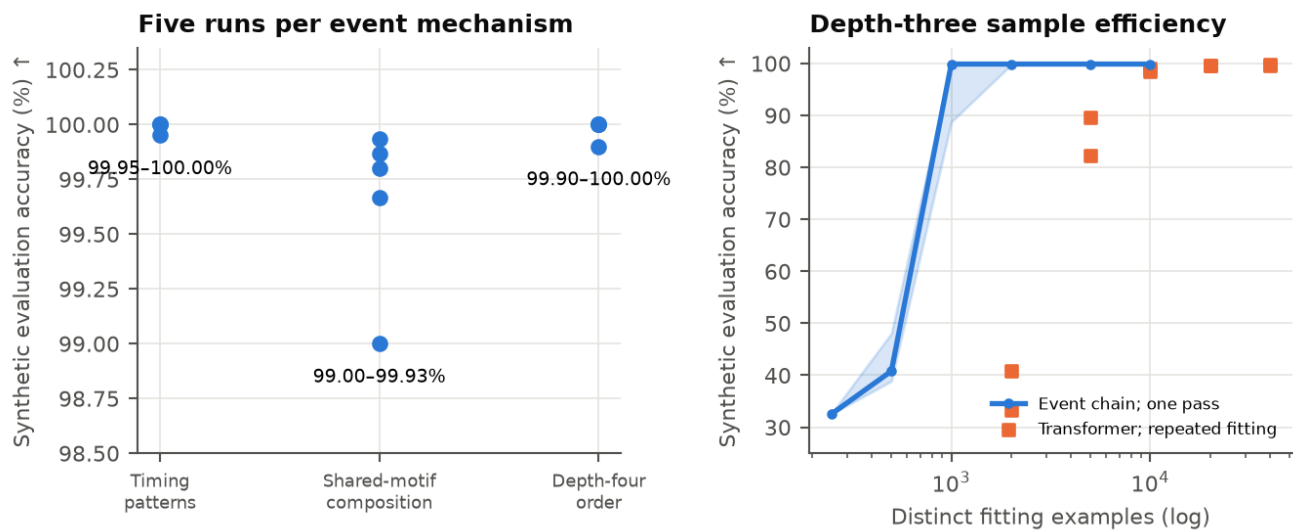
The phase rule costs 188 logical operations per triple; the saved LSTM and Transformer cost 104,518 and 155,592 and score 1.95% and 3.60%. At four times the recall context, the event encoder plus learned pointer scores 100% at 728,602 operations; both compact controls score 7.42% at 2.24M and 4.46M operations. The phase model receives a periodic representation with period 17, and retrieval has a pointer mechanism. These useful priors explain the task advantage and are part of what must transfer to harder tasks.

Learning work is also selective. The periodic teacher makes 29,003 mistaken-example updates and 145,015 learned-scalar update visits. The dense arithmetic controls make 4,800 Adam steps: 92.2M parameter visits for the LSTM and 132.5M for the Transformer. These count parameter updates, excluding optimizer state and backward arithmetic; they are not training FLOPs or joules.

Arithmetic: 1,473 fitting triples, a 200-epoch budget, all 3,440 unseen triples; supplied period 17. The phase-only path stops after 47 passes, when an entire fitting pass makes no updates. Recall: 4,000 pointer-fitting examples plus 512 neural-fitting examples; the dense controls receive all 4,512 examples for eight epochs. Width 32 and two generic layers where present, seed 6, one small dense setting. Work is an analytic logical-operation estimate, including configured vector maps, routers, scans, normalization, clock candidates and pointer search. These inference counts are not measured joules or backward/optimizer counts. Full work definitions appear in the evidence appendix.

A demonstrated advantage: learning temporal structure

Event chains learn timing patterns and compose recognizable parts into ordered structures. The preserved five-run results show accurate recognition and strong sample efficiency. The right panel compares distinct examples rather than incompatible activity counters.



Preserved comparison	Sleeping Machines	Transformer	Fitting protocol
Timing patterns	99.95–100%; five runs	99.60–99.80%; two runs	200k examples once / 1M with relative-time bias
Shared-motif composition	99.00–99.93%; five runs	99.60–99.85%; two runs	40k examples once / 1M with relative-time bias
Depth-four order	99.90–100%; five runs	98.95–99.05%; two runs	Same 40k distinct examples; one pass / 50 passes
Depth-three order at 2k examples	99.73–99.93%; five runs	33.25–40.80%; two runs	One pass / repeated fitting on the same 2k examples

The strongest depth-four comparison has approximately ten times fewer classification errors despite the event learner seeing each fitting example once. The sample-efficiency curve makes the next question concrete: can learned embeddings and broader event representations retain that advantage when the inputs no longer supply known temporal parts?

Completed E35, E34, E53/E54 and E36 files. Ranges describe the recorded runs, not confidence intervals. Architectures and optimization differ; synthetic evaluation sets were reused during research. Event deliveries remain activity measurements. Total arithmetic, backward and optimizer work require a declared ledger; activity divided by dense MACs is not a training-cost or power ratio.

Why asynchronous, sparse computation matters

Persistent local state can retain experience without repeatedly reconstructing a whole history. An asynchronous node updates when useful information arrives. A hard race emits one chosen message. These mechanisms create a path to spending less computation and moving less data per useful answer. On an event-oriented processor, inactive nodes and communication links could remain idle.

Spend the next unit of work where it helps

A larger budget can buy a longer memory, better retrieval, a deeper representation for difficult inputs or more local adaptation. The development rule is to measure the held-out improvement from each choice and allocate work to the most useful one. The phase and pointer results show why an appropriate computation can be much cheaper than a generic dense approximation. The scaling program tests how much of this flexibility survives when that computation must itself be learned.

Architecture	Fitting known sequences	Generating or processing a stream
LSTM	Gates depend on prior hidden state; recurrent work is sequential	Retained hidden state; dense gate updates per token
Transformer	Causal attention permits sequence-parallel fitting	Sequential token generation with a key/value cache; adaptation is possible
Sleeping Machines	Affine event memory supports parallel scans once incoming values/times are known	Retained local state and delayed-message queue; only due arrivals execute

Parallel fitting and sequential generation are compatible. The checked event-memory scan preserves outputs and gradients and gives a 1.68× forward/backward CPU speedup in its recorded audit. The current persistent language driver still executes its message queue sequentially; a sequence-parallel language path needs a matching causal schedule and gradient check.

Parallel recurrent computation also appears in linear attention and selective state-space models. EventSSM already processes asynchronous events with scans. These are important controls. The distinctive hypothesis here is the combination of learned timing, sparse communication, credit to alternatives and independent compute budgets; it must earn its advantage empirically.

Primary FLOPs describe the declared event algorithm, including required vector maps, candidate computation, scans, backward and optimizer updates. Simulator padding and dispatch are separate implementation overhead. Physical power also depends on memory, queues, communication and hardware utilization. Demonstrated work savings and projected power savings are distinguished; total device joules have not yet been measured.

A mathematical foundation for trainable computation

Principle	What it enables
Stable transport through depth	The reversible packet/memory construction preserves conditional value and credit norms under its stated operator and boundary assumptions. Readout visibility, routing and optimization remain separate requirements.
Active communication support	Inputs need causal paths through which to interact. A context channel supplies joint information when sparse packets leave local groups disconnected.
Credit to unrealized alternatives	A losing payload or timing choice can show how a different route would change the outcome, while forward computation remains a hard race.
Periodic state as an isometry	Learned rotations/reflections have unit-magnitude occurrence derivatives. Their composition supports reusable arithmetic instead of a table of observed tuples.
Certified composition	Target-constrained min/max composition of phase errors certifies the fitted modular rule across all 4,913 possible tuples; exhaustive checking confirms it.
Natural supervised credit	Categorical and event likelihoods both credit predicted sufficient statistics minus observations. Silence enters through integrated exposure.
Useful optionality	Reserve consists of distinct, attainable future corrections under a causal work budget. Reachability and transferable learning matter alongside immediate loss.
Statistically useful credit	Expected improvement must overcome the curvature cost of fitting noise. Cross-example teacher agreement separates reproducible correction from raw gradient magnitude.

From mathematics to an engineering discipline

The theory connects representation, topology, clocks and optimization. Expressivity describes what a network can compute; transport describes whether information and credit survive; the objective describes what the teacher asks it to learn. These pieces must agree. The periodic certificate is one concrete case where the formal model explains and verifies a learned computation.

A common implementation makes these principles reusable across tasks. Efficient primitives can own a computation when its structure is known; a deep carrier can learn representations when it is not. The research objective is to combine this flexibility with affordable route discovery and increasingly capable models.

Formal derivations and their assumptions are indexed in the project's theory notes. Conditional stability and a certificate for a fitted rule do not establish global optimizer convergence or a scaling law. The program builds on established deep spiking and sparse conditional computation; its focus is the combination of useful temporal operators, hard causal routing and counterfactual learning. Related survey: Direct training of deep spiking networks.

Potential: what the demonstrated capabilities put within reach

The opportunity is intelligence that learns reusable structure and spends computation in proportion to useful activity. Learned temporal state, specialized retrieval, temporal composition and certified periodic computation already provide working foundations. The shared implementation gives those mechanisms a common place to develop.

Compact prediction and memory

Local predictive memories can support compression, stream forecasting and adaptation to recurring patterns. Learned pointer rules can keep their meaning as context grows. These are useful ingredients for models that retain experience without recomputing an entire dense history at each query.

A reusable event-to-decision module

Sound, event-camera vision, touch and telemetry all arrive as evolving evidence. A capable recognizer could maintain context, identify meaningful patterns and answer as soon as confidence is sufficient. Deep speech learning and temporal composition establish parts of this capability. The new temporal encoder improves private speech accuracy by 7.42 points; its gain also holds on disjoint utterances, and its learned clocks contribute to classification. Reliable early decisions and broader generalization are central development goals.

Training efficiency creates capability

Cheaper updates can buy more data, depth and experimentation from the same budget. Better sample efficiency makes each experience more useful. The exact event-memory scan already reduces audited forward/backward CPU time by 1.68×, preserving the checked predictions and gradients. Local timing and routing teachers provide additional routes to efficient learning.

Industrial and scientific applications

Machines and instruments could maintain local predictive models, recognize changes and adapt from new operating conditions. Laboratories could use event histories to select informative measurements and control experiments. These applications connect fast local responses with longer-term memory, close to the source of the observations.

Potential: a new foundation for frontier models

If the architecture combines frontier predictive quality, reliable deep learning and lower total training and inference cost, it would change the practical recipe for building frontier models. Useful capacity, active computation and learning cost could become more independently controllable.

Useful quality at a lower energy cost

A broad advantage can begin with comparable quality at substantially lower measured training or inference energy. A modest quality tradeoff with a large energy saving can also unlock new applications. Tenfold savings, or larger, would transform feasible deployments and research budgets; these are conditional scenarios, not current measured energy ratios.

The economics of creating intelligence

A fixed power and capital budget could produce a more capable model, more specialized models or more research. Teams constrained by compute could enter new scales and applications. Efficient learning would expand what is feasible, including the size and sophistication of frontier training runs.

Capacity that can remain dormant

Large stores of memories and skills could stay available while only relevant portions participate in a decision. Straightforward situations could resolve with little work; difficult ones could recruit more computation. Cheap search, communication and selective credit would make this a different scaling regime from repeatedly activating an entire dense model.

Autonomy in mobile platforms

Robots, vehicles, phones, wearables and remote instruments could sustain perception, memory and adaptation within a mobile power budget. Local intelligence could retain context continuously, react quickly and learn from experience while reducing dependence on continuous connectivity.

Hardware and infrastructure

If sparse communication and persistent local state determine cost, processors and data centers would increasingly optimize those operations: message delivery, delay queues, memory access, candidate lookup and selective updates. Investment would follow measured useful learning per watt and per unit of capital. The architecture could change which accelerators are most valuable and where capable models can operate.

These larger outcomes depend on demonstrating quality, scaling, retention and total resource cost together. The existing results provide concrete footholds for that research program.

What establishes the larger advantage

The ambition is a common model family whose strongest mechanisms remain useful as tasks, data and capacity grow. Arithmetic and retrieval retain their demonstrated strengths in the consolidated implementation. The strongest language mixture is specialized; the generic backbone still needs to demonstrate competitive learned representations. Character and subword-token budgets must be distinguished.

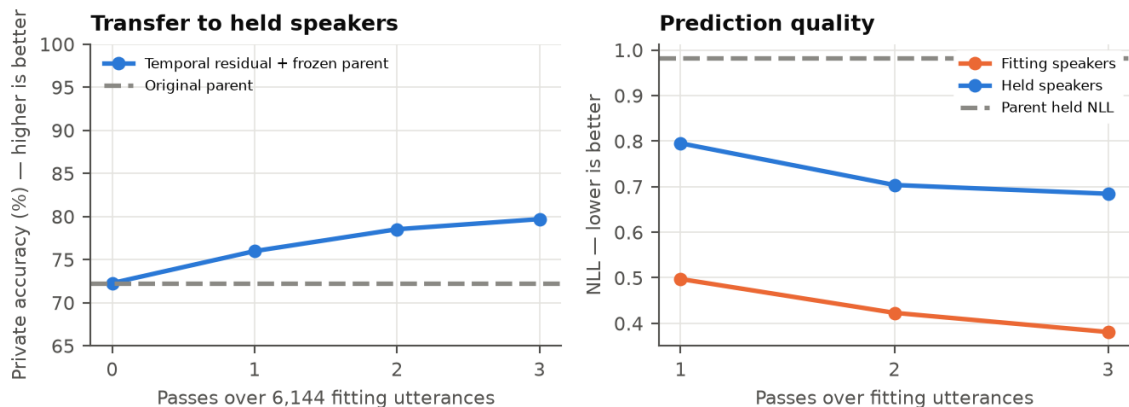
Objective	Decisive evidence
Generic language scaling	Train a learned event backbone that owns the prediction. Scale through declared data budgets with matched Transformer, recurrent and state-space references; record loss, capacity, complete training work, memory traffic, time and joules.
Preserve capabilities	Repeat established generalization and sample-efficiency results within the common model family, with task-appropriate depth and explicit resource accounting.
Strong real-event recognition	Accurate speech and event-camera decisions on complete held-out benchmarks; calibrated confidence and time-to-answer.
Learn routes and representations at scale	Reliable deep credit and useful counterfactual alternatives as width, depth, memory and data increase.
Efficient persistent operation	Maintain local state and pending messages across queries, preserving causal predictions while reducing repeated work.
Lower total energy at useful quality	Measure training and inference joules, memory traffic, latency and communication under declared hardware and quality targets.

Success on these dimensions would turn the current task-level advantages into a broader foundation for frontier models. The project's distinctive resources—timing, local memory, hard selection and credit to alternatives—remain the guide for architecture and learning.

Appendix A. Deep event recognition

The strongest completed speech result in the model family is **79.69%** on 512 private held-speaker utterances (selected single six-block prefix; trained with twelve blocks). Accurate general recognition remains an open capability; published official-test results below are reference targets, evaluated on a different partition.

Private development configuration	Correct	Accuracy $\uparrow$
Original eight-layer parent	370/512	72.27%
Parent + six-block residual; pass 3	408/512	79.69%
Single six-block temporal encoder; pass 0	406/512	79.30%
Calibrated six-block continuation	400/512	78.12%
Bounded twelve-block encoder	400/512	78.12%
Directional twelve-block encoder	401/512	78.32%
Selected single six-block prefix; trained with twelve blocks	408/512	79.69%



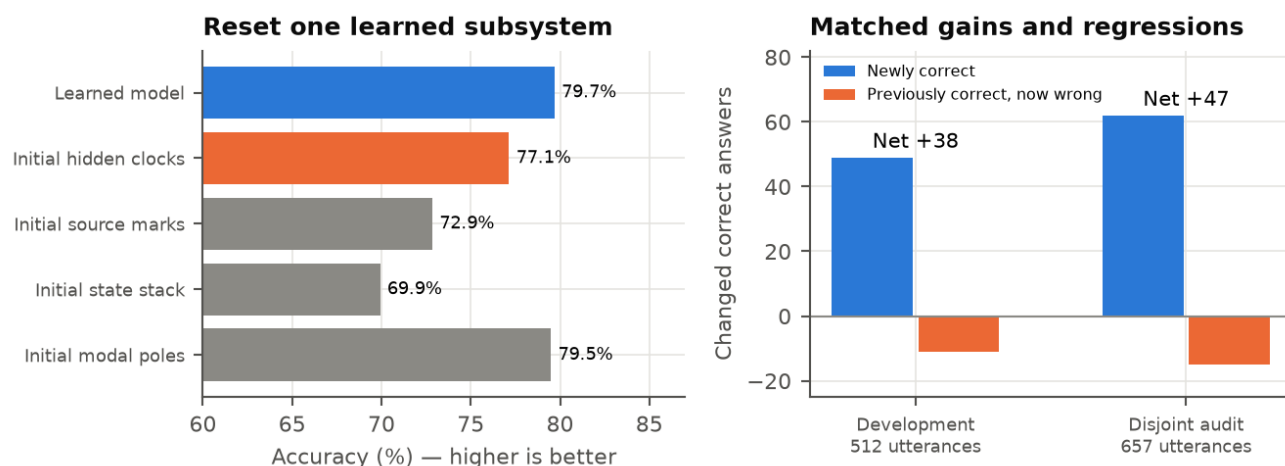
A six-block width-128 temporal encoder learns corrections while the inherited eight-layer parent stays frozen. It adds 395,814 parameters to the parent's 53,296. Signed modal states, nonlinear gates and residual vectors learn from all source identities and original event times before causal pooling. This is a larger parallel model, not fourteen sequential layers; completed-utterance supervision does not yet teach calibrated early answers. On 657 disjoint utterances from the same held speakers, accuracy improves from 70.47% to 77.63%.

Published reference; official-test protocol	Reported accuracy $\uparrow$
EventSSM: asynchronous learned state-space layers	95.9%
S7: input-dependent temporal state	96.3%

Our private sample uses training-file speakers 3/6; official test accuracy is unmeasured. The temporal residual uses three passes and a fresh optimizer; its larger capacity and budget are not a matched single-factor comparison. Its listed score selects the best private-development epoch; the curve shows all three. The original parent was fitted on 4,096 examples. Continuation timers exclude loading; the residual timer includes it. Both include evaluation and are not energy measurements. One seed. References: EventSSM, S7. These published scores were checked against the original papers; they are targets for a full official-test comparison, not scores on our private split.

Appendix A (continued). Learned timing and transfer

Learned delays contribute to the answer. Restoring the hidden clocks to their initial values, with source vectors, state/value maps and the trained classifier retained, loses 13 correct answers. Timing changes the temporal interactions used by the representation; it performs computation.



Same trained readout; one subsystem reset	Correct	Accuracy ↑
All learned parameters retained	408/512	79.69%
Only hidden clocks reset	395/512	77.15%
Only source vectors reset	373/512	72.85%
Only state stack reset	358/512	69.92%
Only decay/frequency parameters reset	407/512	79.49%

The temporal stack and source vectors also learn useful coordinated representations. Resetting the stack loses 50 correct answers; resetting sources loses 35. Decay/frequency resets change one decision. These changes depend on the fitted solution's coordination; their effects cannot be added or treated as a matched comparison of retrained architectures.

The 657-utterance audit is disjoint from fitting and the development sample and uses the selected checkpoint unchanged. It gains 62 correct answers and loses 15, for a net improvement of 47. Both samples use the same two held training speakers. Official-test and additional-speaker generalization are the next evaluation targets.

A stronger mathematical account of routing

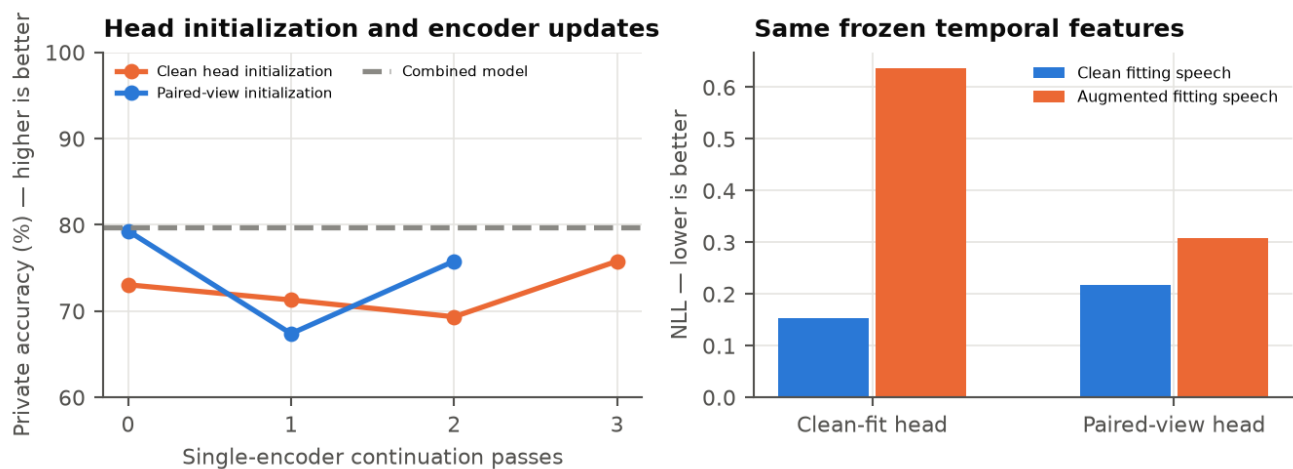
An affine packet summary can preserve both the final state and the average of raw temporal states, including their teachers. This permits richer pooling before expensive nonlinear maps. A second derivation shows why many almost-equal delays may offer little usable choice: their effects point in nearly the same direction. Diverse payloads and temporal modes, sufficient delay spread and downstream visibility determine useful route reserve.

One exploratory run: 449,110 total parameters, inherited parent plus a trained six-block encoder. Three passes use about 33 minutes including preparation/evaluation and 1.79 GiB peak RSS; the guarded host retains at least 10,361 MiB sampled available memory. Complete physical work and joules remain unmeasured. Formal statements and numerical contracts are in THEORY §§226-236.

Appendix A (continued). One temporal encoder

A six-block temporal encoder retains nearly all the combined model's development accuracy through one ordinary query head. Deployment removes the frozen parent: 395,814 parameters replace 449,110. Modal states, gated vector messages and winning delays remain. Its weights inherit earlier encoder training; combined teacher predictions are used only to initialize the head.

Private development configuration	Correct	Accuracy ↑	NLL ↓
Combined model: parent + temporal correction	408/512	79.69%	0.684
Single encoder: clean head, before continuation	374/512	73.05%	0.818
Single encoder: paired head, selected pass zero	406/512	79.30%	0.631
Single encoder: selected trained prefix	408/512	79.69%	0.625



Fitting the head on clean and transformed speech improves held accuracy by 32 answers with the temporal features frozen. Its covariance penalty suppresses class-visible nuisance variation. The right panel compares the heads on the same features; the left shows all subsequent unrestricted encoder passes. The selected trained prefix reaches 518/657 (78.84%) versus 510/657 (77.63%) for the combined model on the reused disjoint audit. Audit labels do not choose the checkpoint.

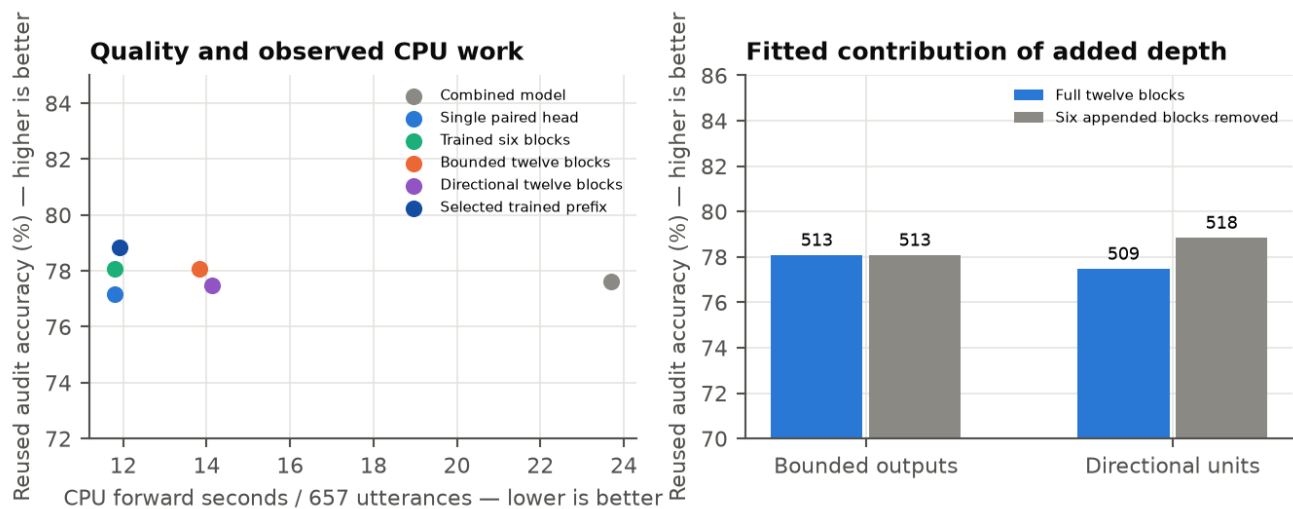
Training the directional twelve-block extension, then selecting its six-block prefix on development, retains the combined model's 408 correct answers with lower NLL and 395,814 deployed parameters. The extra training blocks are removed after their learned contributions reduce held accuracy. This improves deployment quality/work; it does not establish a positive deep-block accuracy gain.

Seed 6, private train-file speakers 3/6; official-test parity remains unmeasured. Head fitting uses 6,144 unique fitting utterances: one clean view for the first arm, clean plus one transformed view for the paired arm. The arms also change regularization and use three/two encoder passes respectively, so total budgets are not matched. All continuation epochs are plotted; selection uses development accuracy, then NLL. Extra teacher/cache/head work and inherited fitting must be charged. Modal/vector maps are locally dense; no empty ticks or event-pair attention are added. The selected prefix additionally inherits the full twelve-block fitting pass; pruning does not erase that training cost. Prefix/full choice is post-hoc private-development selection. Formulae and numerical checks: THEORY §§237–264. One CPU forward evaluation takes 11.90 s for the selected prefix versus 23.71 s for the combined model. Packing/query included, loading excluded; one timing observation, not joules.

Appendix A (continued). Making depth useful

Identity growth preserves the classifier and old teachers while added output maps receive label credit. They must also change useful features. A tightly bounded twelve-block extension learns weights but changes no audit decisions when its six appended blocks are removed.

One matched fitting pass	Parameters	Fit accuracy ↑	Private accuracy ↑	NLL ↓
Six-block control	395,814	90.72%	78.12%	0.639
Twelve blocks: bounded outputs	696,888	90.72%	78.12%	0.639
Twelve blocks: directional units	696,120	90.77%	78.32%	0.644



Directional conditioning normalizes temporal-state features before their output map. An invertible coordinate change rescales classifier-sensitive directions and preserves hidden null directions for later computation. The fixed transform folds into an ordinary map at deployment. Initial outputs and old teachers remain exact; fitting replays verify the actual proposed update.

The directional model reaches 509/657 on the reused audit. Removing its six appended blocks changes 11 predictions: 0 are correct only with the blocks and 9 only without them. This measures fitted contribution with the trained prefix/head retained; it is not a retrained architecture comparison.

All arms inherit the paired-head checkpoint and use 6,144 fitting utterances, the same order, channel/time transformations and one encoder pass. Old-group LR is 0.0000203125; directional new groups use 0.0001953125 from fitting-only replay. Changed normalization and update coordinates form one intervention. New blocks initially add 36 ms latency; labels supervise completed untimed utterances. Audit reuse is explicit; official-test parity remains unmeasured. CPU points are one warmed observation per model, packing/query included and loading excluded; energy is unmeasured. Weight-coordinate folding, all source work and added depth must be charged during training. Local maps remain dense, with no empty ticks or event-pair attention. Theory §§249-264.

Appendix B. Breadth of the common implementation

The common event backbone has development screens across language, event prediction, temporal composition, images and event cameras, in addition to speech, retrieval and arithmetic. These bounded screens establish implementation breadth. Text8 and market rows are hybrids with separately fitted statistical evidence; they are not evidence-free backbone results. The persistent learned language model is reported separately. Task-specific comparison results have their own protocols.

How to read the comparison: accuracy is the percentage of correct answers, so **higher is better**. Bits per character (bpc) and nats/event measure prediction error, so **lower is better**. FLOPs estimate arithmetic work: **lower means less computation**. Each work pair lists the common model first and the Transformer (TF) second.

Task / quality direction	Common quality	TF quality	Inference contractions/query: common / TF ↓	Training total FLOPs: common / TF ↓
Text8 bpc ↓	2.915 bpc	3.729 bpc	1.26M / 1.84M	197.26G / 95.65G
Market event prediction nats/event ↓	3.823 nats/event	4.208 nats/event	1.26M / 1.84M	49.31G / 23.91G
Temporal composition accuracy ↑	97.3%	90.2%	0.29M / 0.38M	25.04G / 14.52G
MNIST accuracy ↑	75.8%	69.9%	1.66M / 2.57M	127.07G / 101.76G
Event-camera gestures accuracy ↑	59.1%	15.9%	23.99M / 124.23M	161.51G / 329.35G

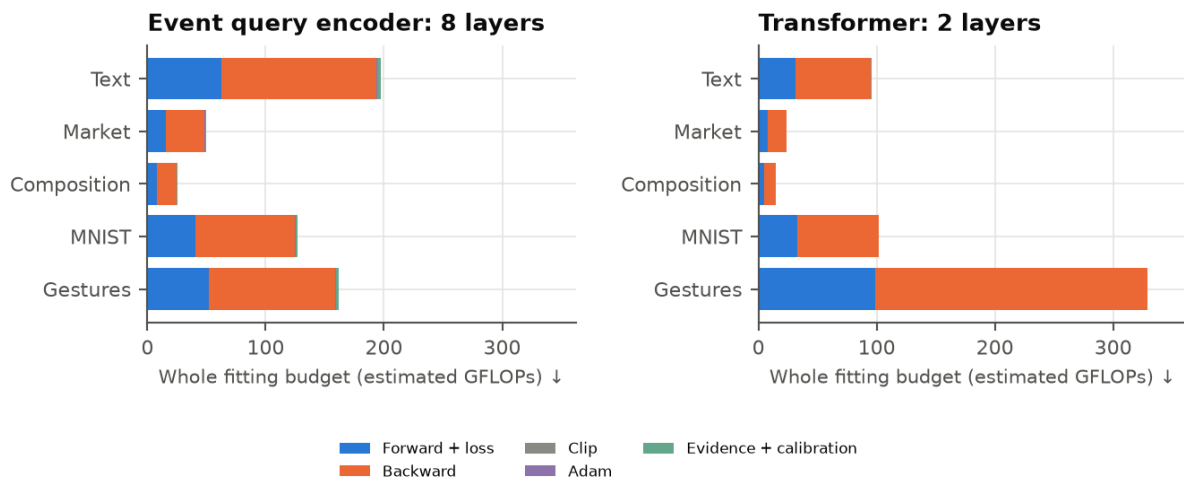
Eight-layer event variants and two-layer Transformer references use width 32, eight epochs and the same neural-fitting/development examples, encoding and objective. Both use batch 16, or batch four for gestures. Depth and capacity differ. M/G/T denote million/billion/trillion. Inference lists unpadded contractions; training covers every fitting epoch, backward, clipping, Adam and event-model evidence/calibration.

Why training can cost more: the race core evaluates all three candidate vector payloads during training, versus only the winner during inference. Its eight layers also exceed the reference's two. With short contexts, that work outweighs the saved attention cost; these rows do not show a training efficiency advantage. On the longer event-camera prefixes, the common model's estimated total uses 49.0% of the reference training arithmetic, including calibration.

Seed 6; neural fit/development counts: text 2,048/256, market 512/256, temporal 1,024/256, MNIST 1,024/256, gestures 88/44. The common text model also has a separately fitted 32,768-character evidence bank; market evidence is fitted on a prior day. The references have no such bank. MNIST uses pooled training-set images; gestures use first-second prefixes and disjoint users. No official real-data test sets are used here. The market fixed-evidence reference is 3.670 nats/event.

Appendix B (continued). Total training cost

The ledger estimates the entire completed fitting budget for each reported model. It includes prediction and loss, backpropagation, gradient clipping and Adam across all eight epochs. The common model also pays for its evidence bank and initial readout calibration. These models were trained from initialization; there is no inherited neural fitting to omit.



Model/task	Forward + loss	Backward	Clip	Adam	Evidence + calibration	Total
Language: Common	62.633	130.873	0.165	0.658	2.930	197.26
Language: TF	31.246	63.990	0.083	0.334	0	95.65
Market: Common	15.657	32.716	0.041	0.164	0.733	49.31
Market: TF	7.811	15.996	0.020	0.081	0	23.91
Temporal: Common	7.725	16.546	0.081	0.324	0.364	25.04
Temporal: TF	4.773	9.540	0.041	0.162	0	14.52
Mnist: Common	40.394	84.376	0.083	0.333	1.886	127.07
Mnist: TF	33.026	68.489	0.049	0.197	0	101.76
Dvs: Common	51.580	107.386	0.028	0.112	2.404	161.51
Dvs: TF	99.023	230.261	0.014	0.057	0	329.35

All table values are estimated GFLOPs for the whole fitting run, not per query. A multiply-add counts as two operations. Forward/loss and backward use the saved E172 four-query operator trace scaled by full-run contractions. The common model uses its logged candidate-map/scan counts. Reference padding is reconstructed from all fitting prefix lengths, the original shuffle seed and batch sizes, including fused attention products. Clipping and Adam are charged once per original step. Other arithmetic and the small calibration eigensolver are estimates.

Event-target arithmetic excludes padding and simulator dispatch/allocation; required candidate maps, losing-value teaching, scans and learning remain charged. References use the same FLOP convention. Evaluation, search, encoding, special functions, integer/index work, comparisons and memory traffic are outside these totals. These single-seed screens have different depths/quality and do not measure event hardware, matched-quality cost or energy. Ledger: `estimate_training_work.py`.

Appendix B (continued). Completed language references

The Transformer and LSTM benchmarks have already been run. Their completed result files remain in the repository and are reused as reference targets for the full learned-event benchmark. Lower bits per character (bpc) means better prediction.

Reference model	Fitting characters / passes	Test bpc ↓	Full training FLOPs ↓
LSTM; width 512, one recurrent layer	10M / six passes	1.799	432.59T
Transformer; width 256, four layers	10M / four passes	1.908	888.78T
LSTM; width 512, one recurrent layer	90M / 6 passes	1.661	3.89P

The 10M references score exactly the same 999,999 text8 targets in [95M,96M), with frozen validation-selected weights and cold initial context. The 90M LSTM uses the same test interval and its saved recurrent scoring protocol. All use the historical 27-character alphabet and 200,000-character validation selection; data budgets, capacities and fitting passes differ.

Training estimates include every fitting step, forward/loss, backpropagation, gradient clipping and Adam. Backward is approximated as twice forward; multiply-add counts as two FLOPs. G/T/P mean billion/trillion/quadrillion. Validation/test evaluation, memory traffic and runtime are outside these arithmetic totals.

Sleeping Machines benchmark status

The learned event-state model is running with 10M fitting characters, four passes, 200,000 validation characters and the same 1M test interval. Its six layers, width 256 and 128 temporal modes have 1,205,805 parameters. Its full test score and training work are pending. The completed 28,403-parameter model's 3.351 development bpc comes from a smaller fitting budget and a different evaluation split; it is not a comparable test result.

Earlier 1M-character references also remain saved: LSTM **2.179** and Transformer **2.367 test bpc**, each with twenty fitting passes. The separate count/copy baseline and the cross-task Transformer/retrieval LSTM comparisons remain in their labeled sections and Appendix B.

Saved evidence: 10M LSTM aligned result; 10M Transformer aligned result; 90M LSTM saved result. An earlier 90M Transformer attempt was interrupted by its RSS watchdog before producing a completed test result; its provenance is preserved.

Appendix B (continued). Separate statistical language baseline

This count/copy predictor does not use the learned Sleeping Machines event backbone. It is a separate statistical system: order-0 through order-6 counts, backoff probabilities and a bounded causal copy cache, combined by learned mixing weights. Its quality/work results must not be attributed to the event architecture.

The statistical predictor and saved neural references score the same 999,999 character targets, starting from a cold context. Parameters are frozen during testing. Earlier observed test characters can supply causal context, including the mixture's bounded 256-character copy cache. Lower bits per character means better prediction.

Predictor	Test bpc ↓	Fitting and selection budget	Estimated training FLOPs ↓
Separate statistical count/copy baseline	1.727	10M count fitting + three 1M mixing-rate trials	4.47G + integer count construction
Count/copy plus causal word context	1.719	10M count fitting + three 1M mixing-rate trials	4.80G + integer count construction

The neural totals charge all original optimizer steps that produced the inherited E174 checkpoints: forward, estimated 2×-forward backward, clipping and Adam. Alignment evaluation is excluded. The statistical floating estimate charges expert probability preparation and all three mixing-rate trials, including local gradients and weight updates. Exp/log/root evaluations count as one operation in these language estimates. Count construction additionally uses about 80M integer count presentations, plus sorting, lookup and hashing; that work is not quantified as FLOPs. The floating totals alone cannot establish total compute, runtime or energy savings.

The count/copy mixture improves on LSTM by 0.072 bpc and Transformer by 0.181 bpc. These results establish useful specialized prediction; generic learned representations are assessed in the separate language screen.

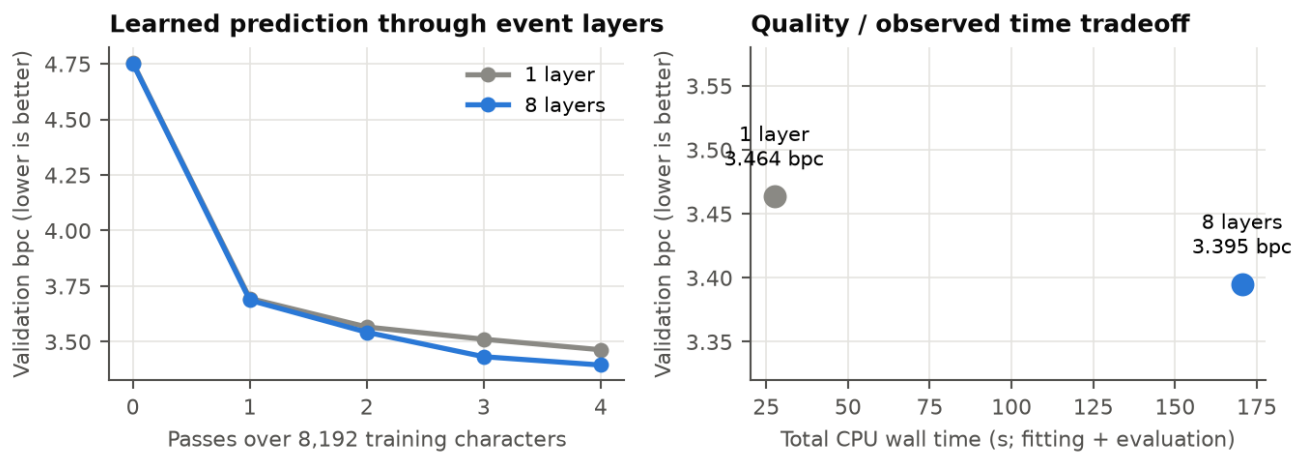
The statistical baseline's count arrays occupy 66.55 MB. Vocabulary, capacities, optimization and fitting budgets differ from the neural references. The comparison does not measure total training energy or a matched-capacity advantage.

All three use the same historical 27-character alphabet. Modern shared subword tokenization is a separate comparison gate for the learned event architecture, described later in this appendix.

One exploratory seed. Text8 offsets: count fitting [0,10M), mixing-weight validation [90M,91M), test [95M,96M); test index zero is excluded for all three predictors. The mixture selects its update rate on validation. Saved neural weights are unchanged. Results: E173/E174; stream contract: E175. At 90M training characters, reference test scores are 1.661 for the LSTM. Estimated training work (forward, backward, Adam and gradient clipping): LSTM: 3.89 PFLOP. Shape-based estimates count multiply-add as two operations; backward is approximated as twice forward. Validation/test inference is excluded. four-layer Transformer reference results are pending. These are single-seed comparisons; capacities and fitting budgets are not matched.

Appendix B (continued). Learned language and depth

A bounded screen trains the common event backbone to predict the next character. Both configurations use width 32, the same 8,192 training characters, four passes, 32-character contexts and 1,024 validation predictions. All eight layers' value, route and memory-time parameters update. Lower bits per character means better prediction.



Depth	Learned parameters	Validation bpc: lower is better	Total CPU wall time
1	7,671	3.464	27.4 s
8	53,430	3.395	170.5 s

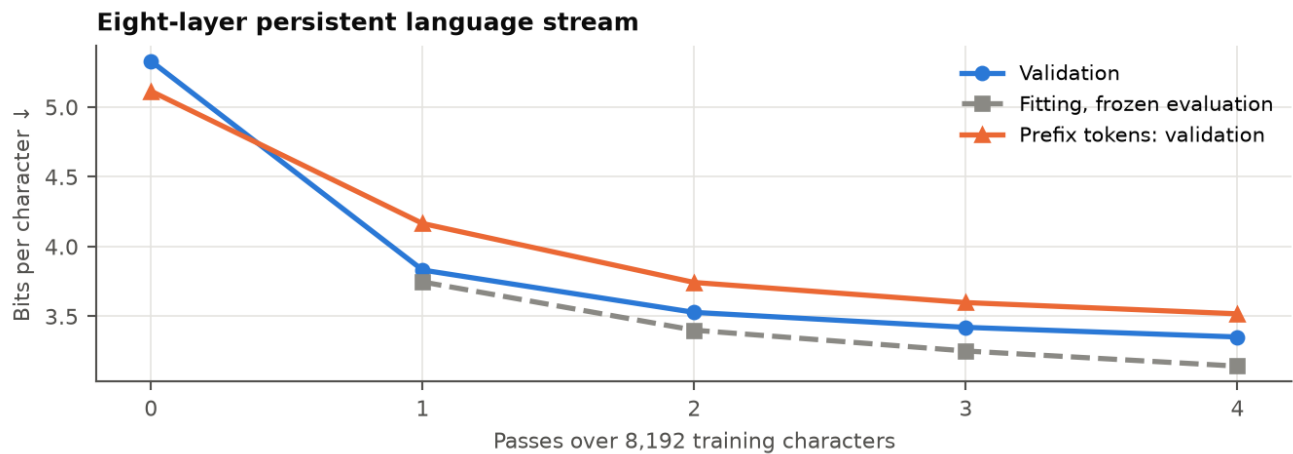
Eight layers improve validation loss from 4.752 to 3.395 bpc, versus 3.464 with one layer. Shuffling preceding characters while preserving the last character, count and timestamps increases the deeper model's loss to 3.805; replacing preceding context raises it to 3.777. These frozen input probes show context sensitivity, not a retrained baseline comparison.

The deeper model has more parameters and takes more CPU time. The quality/time panel includes fitting and evaluation, with backpropagation and optimizer updates executed during fitting. These bounded-query models replay preceding context. The following persistent implementation consumes each character once. Physical memory traffic and joules remain unmeasured.

One seed; different parameter counts. This establishes a generic learned-language foothold and a small depth gain, not competitive large-scale representation, a matched tuned dense-model advantage or a scaling law. The statistical 10M-character mixture remains a separate result. Official test data are untouched. E133 preserves commands, source/data hashes, layer diagnostics and work coverage.

Appendix B (continued). Persistent learned language

The event-state language model consumes each character once and retains local modal memories and its delayed-message queue. Chunk boundaries truncate learning credit without discarding the observed history. Only actual event arrivals evaluate layers; text time is measured in token intervals.



Representation	Parameters	Fitting bpc ↓	Validation bpc ↓	Deliveries/pass
Characters	28,403	3.142	3.351	65,784
Causal prefix tokens	35,163	3.085	3.517	45,816

Validation loss falls from 5.329 to 3.351 bpc. All eight layer teachers are nonzero in every fitting pass. Each pass consumes 8,223 characters including warmup and makes 65,784 block deliveries. The result establishes learning with persistent causal state and no prefix replay.

A train-only 131-token prefix dictionary reduces layer deliveries by 30.4% and observed CPU time by 26.2%, while validation bpc is 3.517. Exact partial-token marginalization scores identical raw targets. Its larger vocabulary fits better but generalizes less well in this small screen: compression alone does not explain or resolve the quality gap.

One seed; 28,403 parameters, width 32, sixteen temporal modes per block. Four passes, 128 Adam steps/pass, credit truncated every 64 characters, 31 warm characters and 1,024 validation targets. Target offsets match the bounded E133 screen; topology, capacity, history and update counts differ, so this is not a matched intervention. Total CPU wall time 364.9 s including fitting/evaluation; peak RSS 364.8 MiB. These are event counts and observed resources, not total arithmetic, physical memory traffic or energy. No official test or large-corpus claim. E176.

Appendix B (continued). Learned event language: training and inference work

These counts belong to the learned eight-layer persistent Sleeping Machines event model that reaches 3.351 validation bpc. It has no count/copy/word experts. Primary counts describe the logical event algorithm; simulator dispatch/allocation is excluded.

Fitting stage	Estimated arithmetic FLOPs
Forward and loss	1.96G
Backward	4.77G
Gradient clipping	43.52M
Optimizer	173.95M
Warmup	7.42M
Total	6.96G

The entire completed budget includes 32,768 fitting targets, 512 Adam/clipping steps and four stream warmups. Special functions add 85.431M evaluations, reported separately from arithmetic FLOPs.

Inference boundary	Per character
Prediction plus NLL-scoring arithmetic	59,741 FLOPs
Additional special functions	1,212 evaluations

A 64-character saved-checkpoint trace measures forward/loss, backward, clipping and warm Adam; the fitting ledger scales those stages by the original 512 steps and separately charges warmup. This is a representative arithmetic estimate, not a whole-run trace. The traced chunk has complete floating-operator formula coverage. Index/queue work, comparisons, memory traffic and evaluation passes are outside the fitting arithmetic boundary.

The 10M-character LSTM/Transformer checkpoints have different data, capacity and achieved quality. Dividing their full fitting budgets by this small run would not establish a fair training advantage. Full learned-event runs must complete before a larger aligned quality/work comparison.

Evidence: `event_language_work/local_event_language_work_20260930T124646Z.json`; quality: E176. Two arithmetic FLOPs per multiply-add. Additional unit-weight special functions would give a different logical-operation total; no event-device runtime or energy has been measured.

Appendix B (continued). Statistical online-learning pilot

The official language comparison freezes parameters during evaluation. A new development-only pilot asks whether causal local learning improves prediction: score each character first, reveal it, then update only the expert mixing weights. Count experts stay frozen and both arms use identical causal copy-cache behavior.

Expert set	Frozen bpc ↓	Online bpc ↓	Extra update FLOPs
Without word	2.605	2.564	2.75M
With causal word	2.537	2.487	3.21M

This pilot reuses a 100,000-character count checkpoint and its learning rate selected on an earlier 2,048-character validation window. It scores 8,191 targets in a fresh 8,192-character development window [90,032,768,90,040,960), excluding the first target. Each adaptive arm performs 8,192 updates, including first-position warmup. Official test data are untouched.

One checkpoint and one window; a specialized adaptive readout, not deep learned TTT or a frontier result. The extra arithmetic column counts only local gradient/weight updates, on top of shared prediction and inherited fitting costs. Total pilot CPU wall time is 0.587 s. Evidence: `online_language/local_online_language_20260930T121741Z.json`.

Compute allocation is the next architectural hypothesis

Independent budgets for active width/depth, dormant capacity, temporal memory, retrieval, credit and adaptation may let extra work buy more prediction quality. The new theory derives conditional marginal-value allocation and retrieval-error bounds, and identifies exposure, routing overhead and hardware utilization as possible limits. A small-scale win does not prove a better scaling exponent or a widening frontier advantage.

Next learned-language comparisons need shared modern subword tokenization, Unicode/byte coverage, suitable rotary/relative position, packed optimized Transformer kernels, sparse MoE and modern recurrent/attention-hybrid controls. Existing event memory already uses relative-time rotations. Token coordinates must remain distinct from learned scheduling delays.

Known sequences can be fitted with input-known affine scans and then generated sequentially with persistent state. Sequence-parallel fitting also permits Transformer adaptation between tokens/chunks. Compare frozen and online arms at identical observations and adaptation budgets; charge cache consistency and update work.

Theory §§280–286 and the frontier compute protocol specify these tests. Modern architecture and larger-scale comparison arms remain proposed, not completed.

Appendix C. Preserved historical results and revisions

Earlier result files remain part of the research record. The following numbers explain older report headlines and why their interpretation changed. They are preserved here with the identified protocol errors; they are not current valid benchmark comparisons.

Earlier statistical language results

Fitting characters	Frozen historical bpc	Online historical bpc
1M	1.808	1.782
10M	1.613	1.593
90M	1.504	1.483

These E79 mixtures use counts, a partial-word expert and a 256-character copy window. The partial-word key depended on whether the target character was a space: changing the unseen target changed the predicted distribution. The old 1.613/1.504 headlines therefore cannot establish a causal language advantage. The corrected E173 10M results are 1.727 without word context and 1.719 with causal word context; a corrected 90M mixture comparison remains open.

Earlier event world-model results

Historical predictor	Day 6 log-likelihood ↑	Day 7 log-likelihood ↑
Event hazard + rate/flow state	-2.184	-1.999
Event hazard + per-type state	-2.159	-1.973
Event hazard + per-type state; finer gap bank	-1.912	-1.734
Saved Transformer Hawkes reference	-1.971	-1.816

The event hazard models use sparse conditional memories and local rate/flow state. Their frozen test parameters and causal state updates are useful mechanisms. However, the event-size threshold was fitted across all seven pilot days, including the evaluation days, for both the event and neural references. Day resets and warmup exclusions also differ. The recorded gap needs fitting-only preprocessing and aligned rescoreing before it supports a held-day advantage.

Source and numerical review; E79 language records; E57 world-model records. Preserved timing, composition, retrieval and modular results remain in the main report. New learned-model benchmarks add evidence; they do not erase these earlier runs.

Appendix D. Evidence and metric definitions

Metric	Interpretation
Bits per character	Held-out negative log probability in base two; lower is better next-character prediction.
Accuracy	Fraction of correct class decisions on the declared development or test protocol.
Event likelihood	Scores both the next event type and waiting time, including the observed silence.
Arithmetic FLOPs	Multiply-add counts as two operations. Tables state whether special functions are separate or assigned unit cost; these conventions must be aligned before forming ratios.
Logical operations	The structured-task ledger assigns 2 units per MAC and 1 per other scalar arithmetic/nonlinear operation or estimated sort comparison.
Resource boundary	Event-target arithmetic includes required active algorithm work. Memory traffic, queues, indexing and simulator overhead are reported separately where measured. Activity counts alone do not determine total FLOPs.
Energy	Measured total joules over an explicit boundary. Operation estimates and CPU timings support work comparisons, but are not joule measurements.

The evidence is preserved in versioned result summaries with configurations, split identities, learning curves and source hashes. E173/E174 support the language comparison; E61 supports retrieval; E34/E53/E54 support native composition; E41 supports the original periodic computation. E121/E124 establish consolidated arithmetic and its certificate; E123 supplies the new dense controls and E124 the operation ledger. E118/E119/E122/E125/E126 support deep speech, readout and causal-context comparisons; E127–E131 audit credit geometry, hard race boundaries and separate key/value learning; E132 checks a joint race-credit formalism, E133 supplies the language/depth screen, and E134–E135 test whole-value credit and content-selective temporal memory. E136 audits reversible augmented transport and its supervised memory boundary, including twelve-layer query/learning interventions. E137 tests compact memory queries and class-visible credit geometry. E138–E141 examine richer source messages and trainable signed temporal memory, with exact local teacher and initial-nesting contracts. E142 establishes signed-state and first-coalescing identities; E143 tests a larger nonlinear temporal residual learner, and E144 audits simultaneous state/query pooling. E171 reproduces the consolidated screens and selected speech answers, and checks causal input boundaries. E172 records complete training-step arithmetic; E175 checks the generic persistent language stream.

The project theory index contains formal assumptions and proofs. Research findings retain detailed analyses and the full experimental record. The model documentation describes reproducible configurations and operational procedures. This report presents the project, its evidence and its potential.